

SECURITY LOG ANALYSIS REPORT

Manual Log Investigation & Threat Detection

Date: April 1, 2026 | Dataset: 1,000 Log Entries

1. Introduction

This report presents the findings of a manual log analysis conducted on a dataset of 1,000 security log entries recorded on April 1, 2026. The logs were collected from multiple hosts within the organization's network, including APP02, DC01, FW01, MAIL01, WEB01, WS100, and WS101.

The objective of this analysis was to identify suspicious activities, detect potential security threats, and provide actionable recommendations to improve the organization's security posture. The analysis was carried out using Microsoft Excel, utilizing filters, sorting, and pivot tables to investigate authentication failures, account lockouts, privilege escalation, and brute-force attack patterns.

2. Key Findings

2.1 Dataset Overview

Total Log Entries	1,000
Date of Logs	April 1, 2026
Unique Event IDs	4624, 4625, 4634, 4672, 4688, 4740
Unique Hostnames	APP02, DC01, FW01, MAIL01, WEB01, WS100, WS101
Total Source IPs	59 unique IPs (10.0.1.x range)

2.2 Most Failed Logins by User

The following users recorded the highest number of failed login attempts:

svc_backup	133	Highest — Service account, very suspicious
charlie	131	High
bob	130	High

alice	126	High
david	126	High
guest	125	High — Guest account should not be active
eve	119	Medium-High
admin	109	High — Admin account targeted

2.3 Account Lockouts

The following accounts experienced lockouts during the observation period:

guest	28
alice	26
charlie	25
david	19
eve	18
bob	16
svc_backup	16
admin	13

2.4 Suspicious IP Addresses

Several IP addresses recorded the highest number of login attempts (28 each):

- 10.0.1.10 — 28 attempts
- 10.0.1.21 — 28 attempts
- 10.0.1.55 — 28 attempts

Other notable IPs include 10.0.1.14 (27), 10.0.1.32 (27), and 10.0.1.26 (26). The concentration of activity within the 10.0.1.x subnet suggests an internal or compromised-internal source of attack.

3. Analysis

3.1 Brute-Force Evidence

Strong evidence of brute-force attacks was observed in the dataset. The clearest example involves the svc_backup account:

- Failed login at 08:09
- Failed login at 08:14

- Failed login at 08:37
- Successful login at 08:37

This pattern of multiple failed attempts followed by a successful login is a classic indicator of a brute-force attack that ultimately succeeded. The guest account also showed a similar pattern: two failed attempts followed immediately by a successful login at 08:14.

The high volume of failed logins across ALL user accounts (109–133 failures per user) further supports coordinated and persistent brute-force activity targeting the entire organization.

3.2 Username Enumeration Patterns

Username enumeration occurs when an attacker systematically tries different usernames to identify valid accounts. Evidence of this was observed in the following:

- All 8 user accounts (admin, alice, bob, charlie, david, eve, guest, svc_backup) were targeted with nearly equal volumes of failed logins (109–133 attempts each).
- The even distribution of attack attempts across all usernames suggests the attacker already had a list of valid usernames or was systematically testing accounts.
- The targeting of the guest and svc_backup accounts (which are often overlooked in security configurations) indicates deliberate enumeration of service and default accounts.

3.3 Password Spraying Patterns

Password spraying involves trying a small number of common passwords against many accounts to avoid lockout thresholds. Evidence observed:

- Multiple user accounts experienced failed login attempts from the same source IPs within the same network range (10.0.1.x).
- Attempts were spread evenly across all 8 users, which is consistent with spraying behavior (rather than targeting one account intensely).
- 59 different source IPs were involved, suggesting either a distributed attack or use of multiple compromised machines.

3.4 Time-Based Attack Patterns

The attack activity began in the early hours of the day and continued across multiple time intervals. Key observations:

- Initial suspicious activity was observed around 08:17, where repeated failed login attempts began for multiple user accounts.
- A successful authentication for the svc_backup account was recorded at 08:37, following multiple failed attempts, indicating a possible brute-force success pattern.
- Privilege escalation events (Event ID 4672) were observed after suspicious logon activity, with some occurring much later in the day (e.g., 23:50), suggesting continued or persistent access.
- The distribution of activity across different timestamps throughout the day indicates that the attack was not a single burst but occurred in intermittent intervals, which is consistent

with automated tools designed to avoid detection.

3.5 Privilege Escalation Analysis

Privilege escalation activity (Event ID 4672 — Special Privileges Assigned) was observed for all users. Critically:

- svc_backup received elevated privileges (132 times) — the most of any account
- guests received elevated privileges (124 times) — highly abnormal for a guest account.
- Suspicious logons were followed by privilege escalation for alice, guest, david, and svc_backup.
- Privilege escalation was observed across all hosts: APP02, WS101, WS100, FW01, WEB01, MAIL01, and DC01.

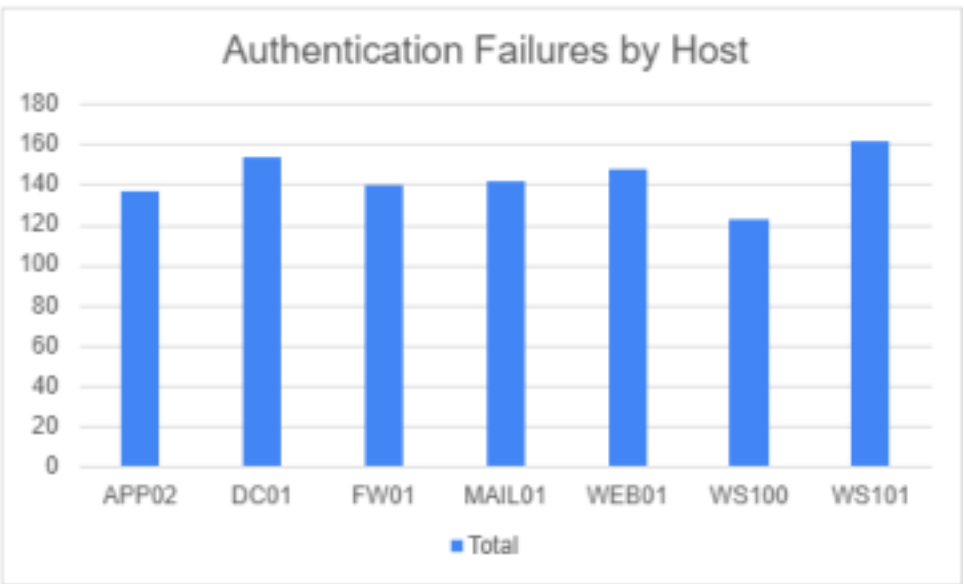
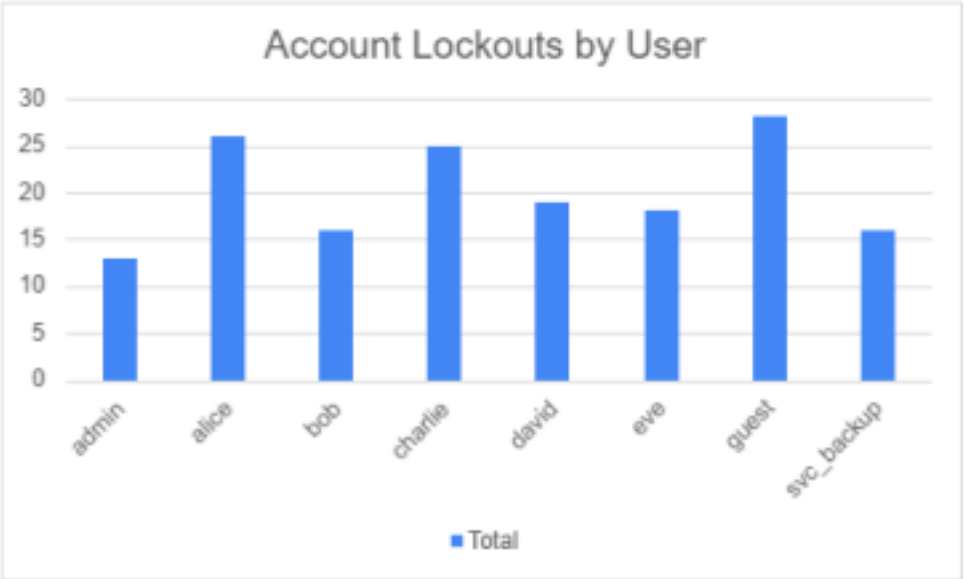
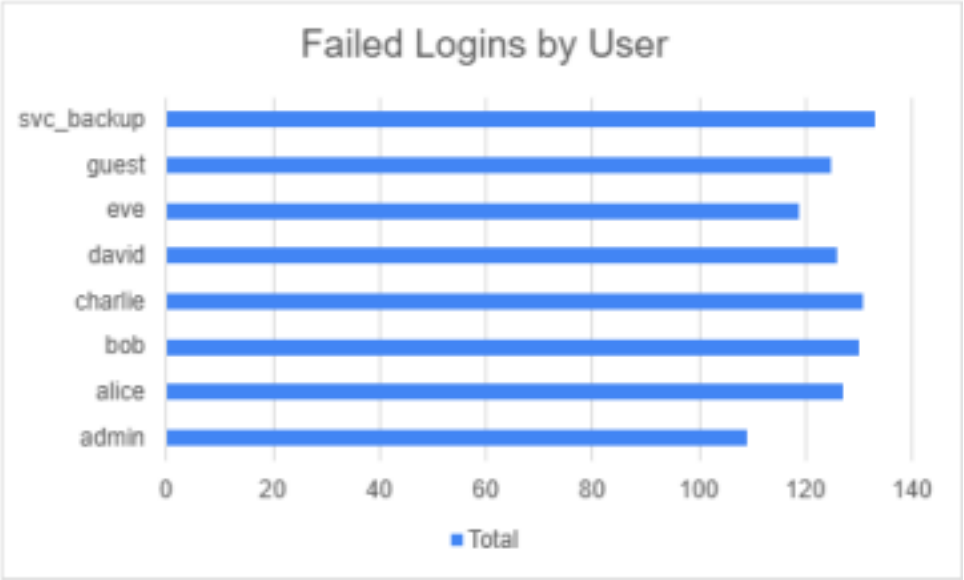
The assignment of elevated privileges to accounts such as guest and svc_backup following suspicious login activity suggests potential misuse of privileged access and warrants further investigation.

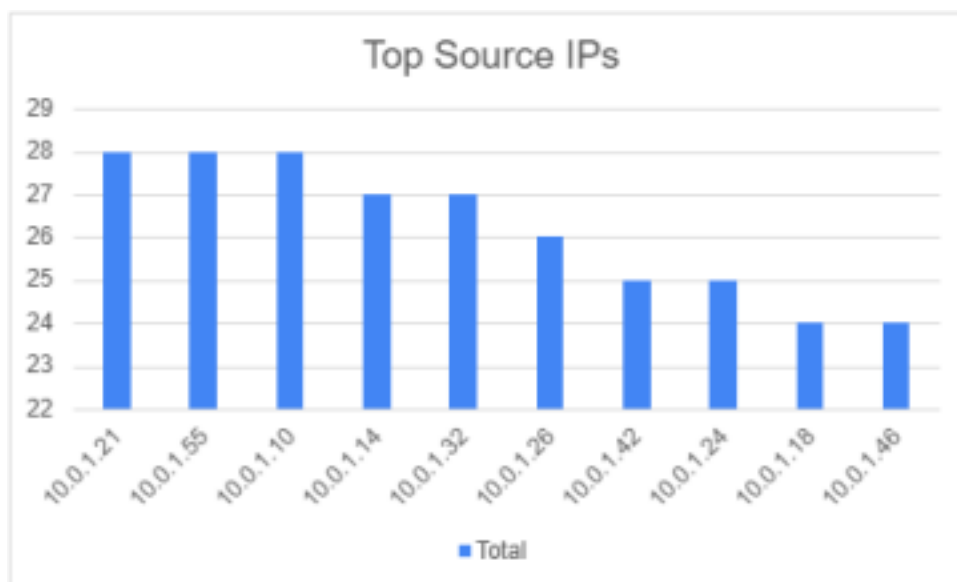
3.6 Suspicious IP Deep Dive — 10.0.1.10

IP Address	10.0.1.10
Total Attempts	28
Unique Usernames Tried	Multiple (admin, alice, bob, charlie, david, eve, guest, svc_backup)
Time Range	08:26 to 00:52 the following day (intermittent activity across multiple sessions)
Risk Level	HIGH

This IP made repeated login attempts across multiple user accounts over an extended period (08:26 to 00:52 the following day), indicating likely automated brute-force or password spraying activity

4. Visual Insights





5. Recommendations

Implement Account Lockout Policies:

Configure account lockout policies to automatically lock accounts after 3–5 consecutive failed login attempts. This will help slow down or prevent brute-force attacks. Ensure the lockout duration is sufficient (e.g., 15–30 minutes) to deter automated tools

Disable or Restrict Default and Guest Accounts

The *guest* account should be disabled immediately, as it has no legitimate business need and was heavily targeted. The *svc_backup* service account should be reviewed, with privileges restricted to only what is necessary, and interactive logins should be disabled where possible.

Implement Multi-Factor Authentication (MFA)

Enable MFA for all user accounts, especially privileged accounts such as *admin* and *svc_backup*. Even if an attacker obtains valid credentials, MFA adds an additional layer of security to prevent unauthorized access.

Deploy Intrusion Detection and SIEM Alerting

Configure real-time alerts for:

- Multiple failed login attempts from a single IP within a short timeframe
- Successful logins following repeated failures (possible brute-force patterns)
- Privilege assignment events (Event ID 4672), especially for

Investigate and Monitor Suspicious IPs

The IP addresses *10.0.1.10*, *10.0.1.21*, and *10.0.1.55* should be investigated. If these IPs belong to internal systems, they may indicate compromised hosts. Network segmentation, monitoring, and access controls should be applied to limit potential lateral movement.

The IPs *10.0.1.10*, *10.0.1.21*, and *10.0.1.55* should be investigated immediately. If they belong to internal machines, those systems may be compromised. Consider network segmentation and IP-based blocking for repeated offenders.

Conclusion

This log analysis revealed significant and concerning security activity within the organization's network on April 1, 2026. The dataset of 1,000 log entries provided clear evidence of:

- Coordinated brute-force attacks targeting multiple user accounts, with *svc_backup* showing a successful login following multiple failed attempts.
- Password spraying patterns distributed across 59 source IPs and all 8 user accounts.
- Possible username enumeration, suggesting systematic probing of valid account names.
- Suspicious privilege escalation activity following login attempts, particularly for the *guest* and *svc_backup* accounts.
- WS101 identified as the most heavily targeted host, recording 160 authentication failures.

The evidence strongly suggests a coordinated and automated attack pattern, with potential partial success on targeted accounts. Immediate action is required to contain the threat, investigate affected accounts, and implement the security measures recommended in this report.

The organization should treat this as a potential security incident until all affected accounts have been audited, passwords reset, and the source of the activity fully investigated.